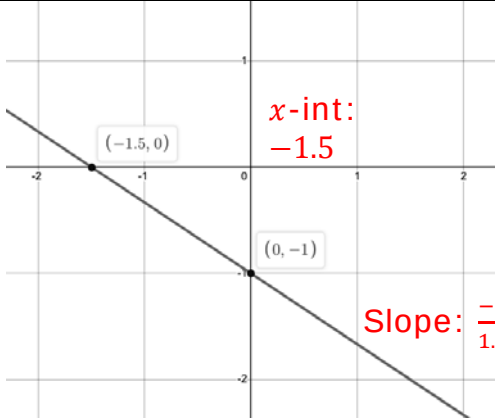


For questions 1 through 4, use the table showing Lines A and B.

Line A	Line B
$B(x) = -\frac{2}{3}x + 3$ Slope: $-\frac{2}{3}$ x-int: $0 = \frac{-2}{3}x + 3$ $\frac{-3}{2} \cdot \frac{-3}{1} = \frac{-3}{2} \cdot \frac{-2}{3}x$ $\frac{9}{2} = 4.5 = x$	 x-int: -1.5 Slope: $\frac{-1}{1.5} = -\frac{2}{3}$

- Which linear function has the larger slope?
The slopes are the same.
- Which line has the larger x -intercept?
Line A
- Name all the key feature(s) that are the same for both linear functions.
Both linear functions have the same slope, domain, and range. They are also both decreasing.
- Create a linear function that has the same y -intercept as Line A.
Answers may vary. $C(x) = \frac{1}{4}x + 3$
- Complete the statements below to describe the end behavior of Line A.

a. As $x \rightarrow \infty$, $y \rightarrow$

☒ $-\infty$
☐ ∞

b. As $x \rightarrow -\infty$, $y \rightarrow$

☐ $-\infty$
☒ ∞

For questions 6 through 10, use the table showing Lines A and B .

Line A	Line B										
x -int: $0 = 2x + 6$ $-6 \quad -6$ $-3 = x$ A line passes through the points $(1, 9)$ and $(4, 15)$. Slope: $\frac{15-9}{4-1} = \frac{6}{3} = 2$ $y - 9 = 2(x - 1)$ $y - 9 = 2x - 2$ $+9 \quad +9$ $y = 2x + 7$ y -int: 7	x -int: -15 Slope: $\frac{7-6}{6-3} = \frac{1}{3}$ <table border="1"> <thead> <tr> <th>x</th><th>y</th></tr> </thead> <tbody> <tr> <td>-3</td><td>4</td></tr> <tr> <td>0</td><td>5</td></tr> <tr> <td>3</td><td>6</td></tr> <tr> <td>6</td><td>7</td></tr> </tbody> </table> y -int: 5	x	y	-3	4	0	5	3	6	6	7
x	y										
-3	4										
0	5										
3	6										
6	7										

6. Which linear function has the smaller slope?

Line B

$$y - 7 = \frac{1}{3}(x - 6)$$

$$y - 7 = \frac{1}{3}x - 2$$

$$+7 \quad +7$$

$$y = \frac{1}{3}x + 5$$

7. Which linear function has the larger y -intercept?

Line A

8. Name the key feature(s) that are different for the two functions.

Line A and Line B have different x - and y -intercepts and slopes.

9. Create a linear function that has the same slope as Line B.

Answers may vary. $D(x) = \frac{1}{3}x - 6$

10. Consider the end behavior for the two functions.

- a. Complete the statement.

Lines A and B have

- ☒ the same end behavior.
☐ different end behaviors.

- b. Write the end behavior of the functions.

As $x \rightarrow \infty$, $y \rightarrow \infty$.

As $x \rightarrow -\infty$, $y \rightarrow -\infty$.